INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Spring Semester Examination 2023-24

Date of Examination: 18/04/2024 **Session: (FN/AN)** **Duration:** 3 Hrs **Full Marks:** 50

Subject No. : CE 31004 **Subject :** Design of Steel Structures

Department/Center/School: Department of Civil Engineering

Specific charts, graph paper, log book etc., required

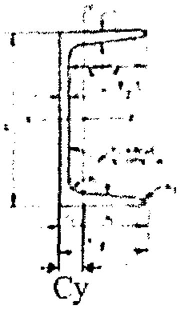
Special Instructions (if any) :

- **This exam paper contains 3 pages + 1 sheets of codal provisions (total 4 pages). There are 7 questions in pages 1, 2 and 3, and helpful information on page 4. Check to see if any pages are missing.**
- **You are not allowed any books or notes. Relevant portions of IS Code are provided at the end of the question paper on page 4**
- **All Questions are compulsory.**

USE SUITABLE ASSUMPTIONS WHEREVER NEEDED. State your assumptions.

Question 1 [5]

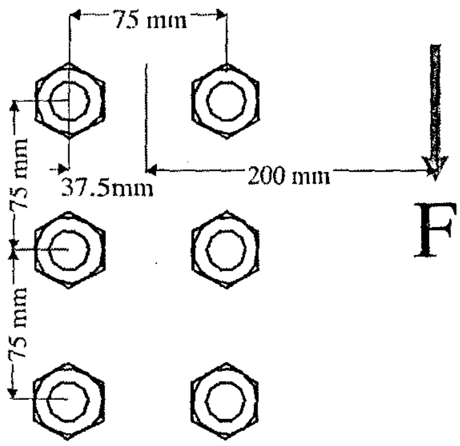
Calculate the design compressive force capacity for ISMC 300 column with effective length 5 m and simply supported boundary conditions, under dead and live load combination. Column section properties are as follows. Assume steel grade as E350.



Name	Area [mm^2]	Cy [mm]	Ix [cm^4]	Iy [cm^4]	rx [mm]	ry [mm]
ISMC 300	4564	23.6	6362.6	310.8	118.1	26.1

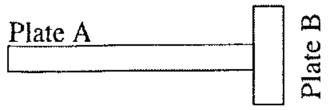
Question 2 [5]

Calculate the maximum shear force *F* that can be resisted by the bearing-type bolt group shown below. Assume bolt diameter as 16 mm, bolt ultimate strength as 400 MPa, and threads are not in the shear plane. Ignore the connecting plate.



Question 3 [5]

Plate A is welded with Plate B using a double-sided fillet weld intermittently (6 mm weld length followed by 3 mm gap). The size of the weld is 12 mm. Draw the figure below and label the weld with appropriate details as per standard convention.

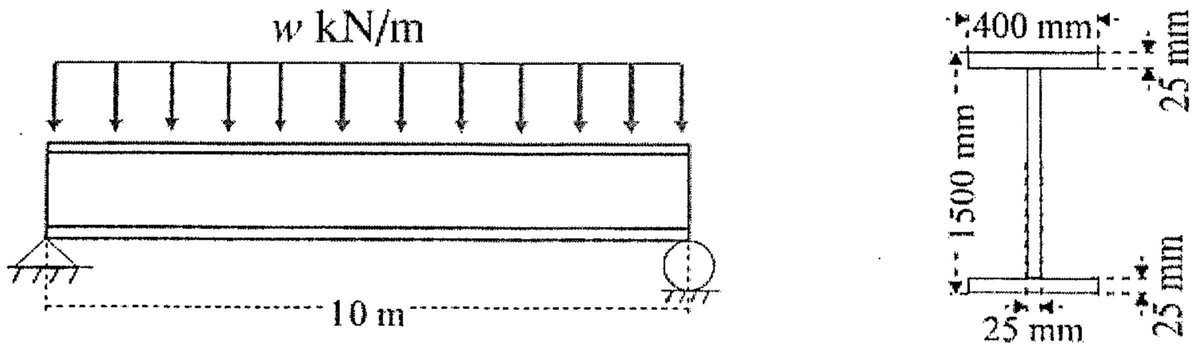


Question 4 [5]

Consider a simply supported column of length L, moment of inertia I and modulus of elasticity E, subjected to a concentric axial compressive load P. Derive the critical value of P at which elastic buckling occurs.

Question 5 [5]

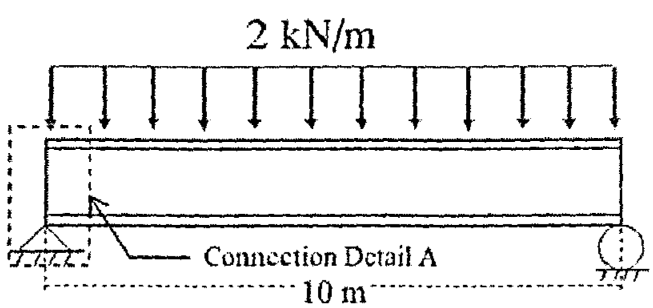
Calculate the design live load carrying capacity w for a simply supported plate girder of length 10 m as shown below (without stiffeners). Assume that the flanges are welded to the web using CJP welds, steel grade is E250, and the girder is continuously braced laterally.

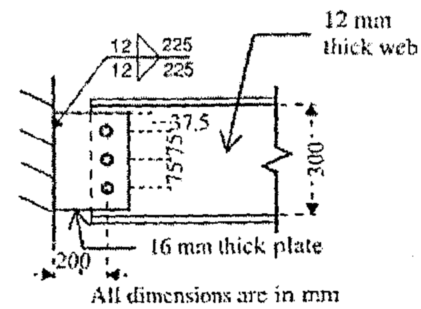


Questions 6 & 7 are on page 3

Question 6 [10]

In the beam shown below, calculate the support reaction due to the applied loading at connection detail A. Determine if connection detail A is adequate to resist this reaction force. Assume bearing type bolts, 12 mm bolt diameter in standard holes, and threads are not in the shear plane. Assume yield stress as 250 MPa and ultimate strength as 415 MPa for all material. List all possible limit states of strength for this connection.





Connection Detail A

Question 7 [15]

(a) Draw a plot to show the variation of Euler buckling load with slenderness ratio. On the same plot, show the range of test results obtained by compressive load experiments on actual columns. Explain why experimentally obtained results are different from Euler buckling load [4]

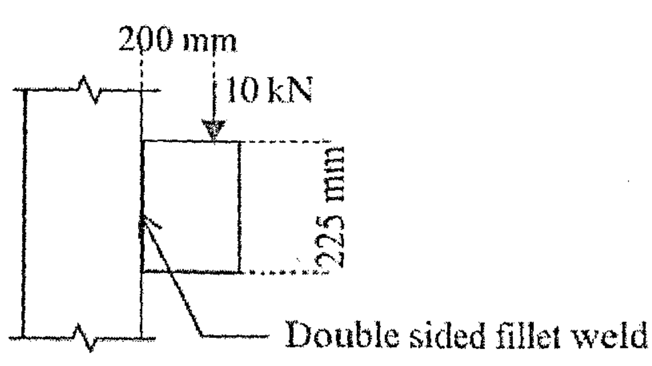
(b) A built-up column is constructed using two ISMC 300 sections such that the radius of gyration is same in both directions. Using the section properties given in Question 1, determine the spacing between the two ISMC 300 sections, and draw the configuration of the built-up section. [3]

(c) In built-up columns, explain why moment needs to be considered during design of batten connection, but there is no moment at lacing connections [2]

(d) Name two types of end restraints (boundary conditions) that need to be checked for unbraced beams [1]

(e) What is a "groove weld"? What are the two types of groove welds commonly used in steel structures? [2]

(f) The plate shown below is connected to the column through a double-sided fillet weld of size 12 mm. If the applied load is P = 10 kN, calculate the vertical shear, transverse shear and resultant shear on the weld [3]



7.1.2 The design compressive strength P_d, of a member is given by:

$$P < P_d$$

where

$P_d = A_e f_{cd}$

where

A_e = gross area for semi compact, compact or plastic sections

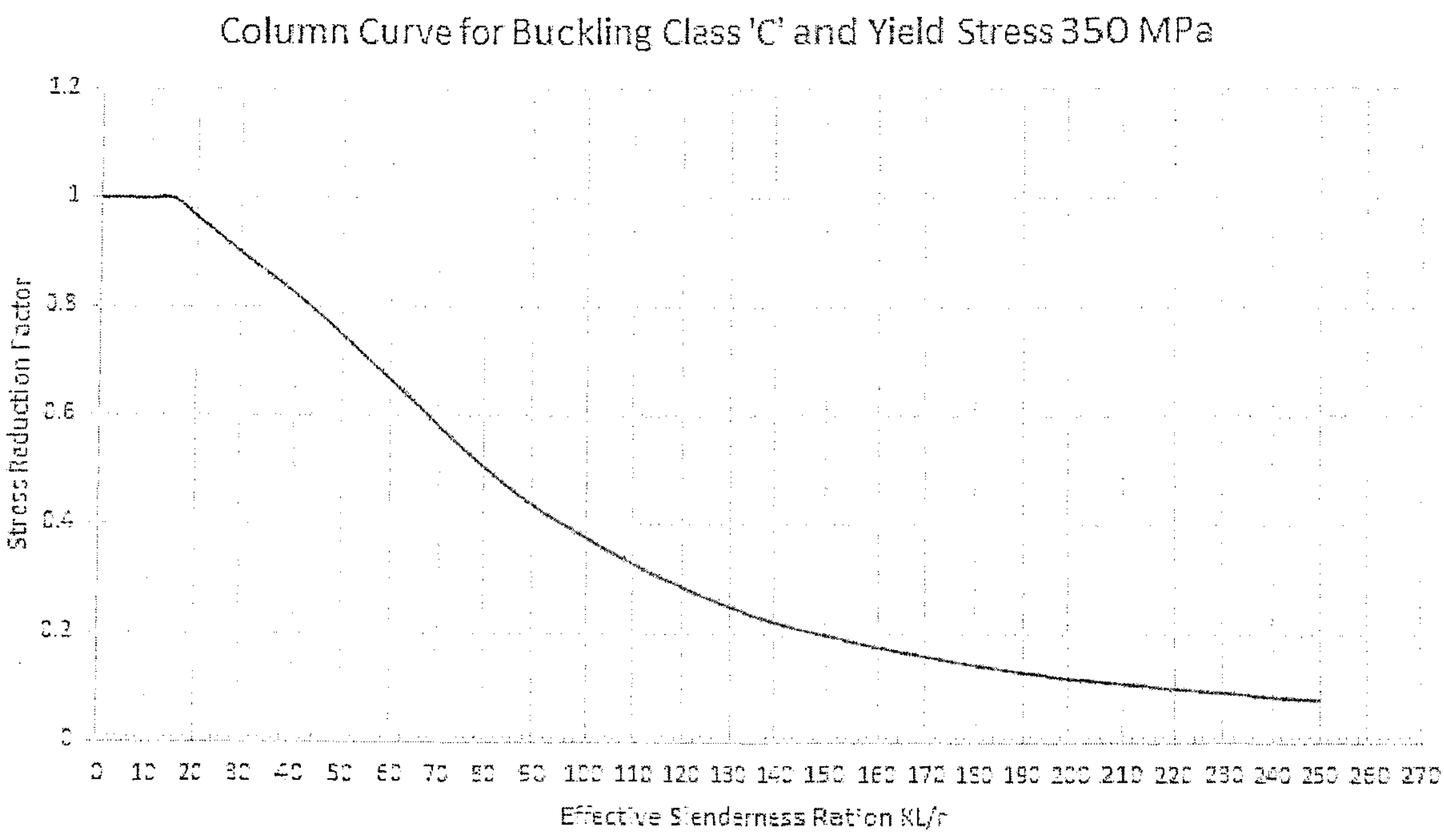
f_{cd} = design compressive stress, obtained as per **7.1.2.1**.

7.1.2.1 The design compressive stress, f_{cd}, of axially loaded compression members shall be calculated using the following equation:

$$f_{cd} = \chi f_y / \gamma_{m0} \leq f_y / \gamma_{m0} \quad \text{where } \gamma_{m0} = 1.1$$

χ = stress reduction factor, see plot below for buckling class 'C' (includes Channels, Lacings and Built-up sections), f_y = 350 MPa and different slenderness ratios

KL/r = effective slenderness ratio or ratio of effective length, KL to appropriate radius of gyration, r;



10.3.3 *Shear Capacity of Bolt*

The design strength of the bolt, V_{dsb} as governed shear strength is given by:

$$V_{dsb} = V_{nsb} / \gamma_{mb}$$

V_{nsb} = nominal shear capacity of a bolt, calculated as follows:

$$V_{nsb} = \frac{f_u}{\sqrt{3}}\left(n_n A_{nb} + n_s A_{sb}\right)$$

f_u = ultimate tensile strength of a bolt;

n_n = number of shear planes with threads intercepting the shear plane;

n_s = number of shear planes without threads intercepting the shear plane;

10.3.4 *Bearing Capacity of the Bolt*

The design bearing strength of a bolt on any plate, V_{dpb} as governed by bearing is given by:

$$V_{dpb} = V_{npb} / \gamma_{mb}$$

where

V_{npb} = nominal bearing strength of a bolt $= 2.5\, k_b\, d\, t\, f_u$

Assume $k_b = 1.0$

d is bolt diameter

t is plate thickness

8.4.1 The nominal plastic shear resistance under pure shear is given by:

$$V_n = V_p$$

where

$$V_p = \frac{A_v f_{yw}}{\sqrt{3}}$$

A_v = shear area, and

f_{yw} = yield strength of the web.

8.4.2 *Resistance to Shear Buckling*

8.4.2.1 Resistance to shear buckling shall be verifi as specified, when

$$d/t_w > 67\varepsilon \text{ for a web without stiffeners}$$

Strength due to net shear rupture

$(0.9 A_{vn} f_u / (\sqrt{3}\, \gamma_{m1})$